BMW Sales Trend Analysis

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Emerging Technologies, Data and Business Analytics - Assignment

1. Introduction and Problem Framing

1.1 Business Context and Problem Definition

The automotive industry is known for its intense competitive nature, and manufacturers must remain continuously up to date with market trends and customers' preferences to optimise their product offerings. For a multinational automotive manufacturer such as BMW, it is vital to have a thorough understanding of their customer preferences and market trends to make suitable strategic decisions that attract and satisfy their customers. This study aims to point out BMW's customer demand by analysing their sales performances through historical sales data from 2010 to 2024.

1.2 Stakeholders

BMW product managers, regional sales managers, and marketing teams could be the primary stakeholders who might hold a strong interest in this analysis. Since product and regional sales managers can have the insights to determine which vehicle characteristics and market zones drive sales performance. While the marketing team can tailor promotional strategies based on consumer preferences. Additionally, external stakeholders such as competitors may also ultilise this finding to evaluate BMW's strengths and weaknesses in the market.

1.3 Analytical Variables

Independent Variable:

- Sales_Volume

Dependent Variables:

- Model
- Engine_Type
- Fuel_Type
- Color
- Transmission
- Price
- Region

1.4 Assumptions

There are several assumptions that can be made during the analysis of the BMW sales trend.

- The data set from Kaggle is realistic and correctly reflects market conditions,
- Sales_Volume records accurately represent consumer demand within the automotive market.
- Variables such as prices, model, region, and fuel type are the key factors that strongly influence customer purchasing decisions.
- Observations in the dataset represents an independent sales record and is not influenced by other entries
- The relationships identified through statistical analysis are assumed to indicate correlations between variables instead of direct causal relationships.

2. Data Cleaning and Preparation

2.1 Dataset Overview

The dataset used for this study was collected from Kaggle, which holds BMW sales records from 2010 to 2024. It consists of 50,000 observations with 11 variables that describe vehicle characteristics, market, and sales attributes. Moreover, the dataset contains a mixture of variable types, such as categorical, date, and numerical variables. Therefore, this variable combination allows for a clear analysis of factors influencing BMW's sales performance.

2.2 Data Exploratory Summary Statistic

Sales Volume by Year				
Year	Min of Sales_Volume	Max of Sales_Volume	Average of Sales_Volume	Sum of Sales_Volume
2010	100	9998	5085	16933445
2011	105	9999	5113	16758941
2012	101	9998	5028	16751895
2013	103	9998	5071	16866733
2014	106	9994	5062	16958960
2015	111	9994	5066	17010207
2016	109	9995	5039	16957550
2017	100	9999	5052	16620811
2018	108	9996	5007	16412273
2019	100	9998	5068	17191956
2020	103	9990	5089	16310843
2021	100	9992	5007	16884666
2022	102	9999	5153	17920946
2023	104	9996	5054	16268654
2024	101	9999	5115	17527854
Grand Total	**100**	**9999**	**5068**	**253375734**

The figure above summarises BMW's sales volumes throughout the 2010 and 2024 period. The results indicate a stable average sales volume over the period, with values hovering around 5,000

units per record. Similarly, the overall sales are consistently present during early periods, with a noticeable upward trend toward the end.

2.3 Data Quality Evaluation and Limitations

Even though the dataset is large and provides varied insights into the demand for BMW cars across different regions. However, the sales volume records display a high level of uniformity and synchronisation, with the values being capped within a range of 100 to 999 units. This pattern suggests that the dataset could possibly be generated by AI, thus constraining its ability to precisely represent actual market behavioral demand.

3. Statistical Analysis

3.1 Preliminary Insights

In general, the observations suggest that BMW sales performance remained stable throughout the 2010 to 2024 period. Although BMW vehicles are relatively expensive, with an average price of approximately $76,000 and an estimated variation of $26,000 *(Appendix C.2).* However, the company is still able to maintain an average sales volume of about 5,000 units. This indicates that BMW has a consistent sales performance, despite variations in pricing *(Appendix C.2)*. Currently, Asia has the largest sales observation, with South Africa being the lowest, although the differences between regions are insignificant *(Appendix C.4)*.

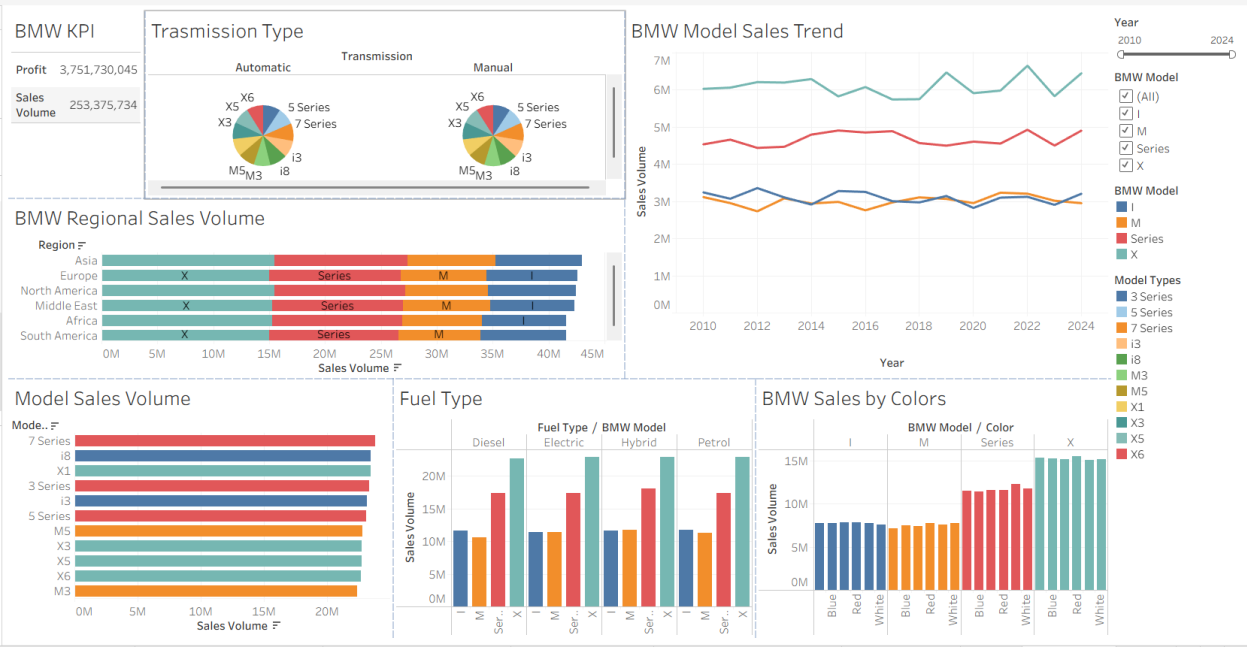
3.2 Correlation and Regression Analysis

Car purchasing decisions are driven by numerous factors, consisting of vehicle type, price, and individual preferences. Therefore, a multiple regression model was applied to examine the combined effects of these variables on sales performance (Leon, 1998; Smalheiser, 2017). On the downside, the model indicates significantly low explanatory power, with an R-squared value close to zero *(Appendix D.1)* (Leon, 1998; Smalheiser, 2017). Additionally, high P-values suggest that the variables are not statistically significant for predictors, while large standard errors suggest unreliable coefficient estimates *(Appendix D.1)* (Leon, 1998; Smalheiser, 2017).

Consequently, the model has limited predictive ability and does not present causal relationships but only reflects weak statistical associations between variables (Leon, 1998; Smalheiser, 2017). This means that vehicle characteristics alone may not sufficiently explain the influence of sales performance. The absence of external factors that can influence purchasing decisions, including consumer income, buying purpose, and brand perception, could be potential limitations. Hence, poor insights are being delivered by the model that truly reflects drivers of BMW sales, and it requires a cautious interpretation.

4. Data Visualisation and Interpretation

4.1 Dashboard Overview & Key Insights



The dashboard outlines BMW's sales performance across models and key attributes over time. The X models generate the highest sales volume, likely due to their selection varieties, followed by the Series models, while the I and M models present the lowest sales volume with fewer model types. When drilled down, the 7 Series and i8 are the most popular individually.

Although sales trends demonstrate a general upward pattern for most models, only M models showcase a slight decline toward the end. Regionally, Asia remains the leading market for BMW sales, even though Europe demonstrates strong growth with the possibility to overtake in the future. In contrast, sales across transmission, fuel type, and color are relatively similar, indicating an insignificant impact on sales performance. Thus, it informs the BMW managers to focus on broader market strategies rather than individual product elements. Even though the visualisation generates crucial insights, it still presents a risk of misinterpretation due to data aggregation, which can result in misleading decision-making.

5. Reflection and Business Insights

5.1 Analytical Decision-Making Process

Initially, a business problem was framed, and exploratory analysis was done to discover the sales patterns. Descriptive statistics and pivot tables were exploited to summarise trends, along with regression to assess relationships of multiple variables and visualisation to aid clearer visual

insights and enhance interpretability, detailed models were grouped into broader categories (Series, I, M, and X) (Andrienko & Andrienko, 2010).

Although the dataset is huge and structured, it is short on key external variables. Additionally, the uniform distribution of sales values indicates the likelihood of data artificiality. As a result, the insights can only be used for reference.

5.2 Ethical, Privacy and Fairness Considerations

The dataset does not hold personal or sensitive information of the actual business, so privacy concerns can be ignored. However, it's potentially artificially synchronise which may lead to misleading results, raising ethical concerns significantly. Moreover, the absence of customer demographics and behavioral factors may lead to bias. Hence, insights should be interpreted carefully and not fully relied upon for decision-making.

5.3 AI Considerations

AI was utilised to assist in generating codes for data loading, cleaning, wrangling, and running analysis models. Furthermore, it is also used for consultation on applying the suitable analytical method, visualisation, and guidance on enhancing interpretability through data wrangling.

5.4 Business Recommendations & Potential Risks

According to the analysis, BMW should concentrate on broader markets strategies instead of focusing on individual products attributes as they exhibit a weak influence on sales performance. The priorities should be placed on expanding model selection, while strengthening market presence in regions displays growing demand such as Europe.

Conversely, the absence of external factors such as market conditions and consumer behaviors are the biggest limitations of the dataset, which raises implementation risks. The incomplete dataset may lead to misleading decisions if applied directly. As a result, these insights should be fostered with additional real-world data and market research before executing.

6. Reference:

Andrienko, G., & Andrienko, N. (2010). A General Framework for Using Aggregation in Visual Exploration of Movement Data. *The Cartographic Journal*, *47*(1), 22–40. https://doi.org/10.1179/000870409x12525737905042

Leon, A. C. (1998). Descriptive and Inferential Statistics. *Www.sciencedirect.com*, *3*, 243–285. https://doi.org/10.1016/B0080-4270(73)00264-9

Smalheiser, N. R. (2017). *Data Literacy : how to make your experiments robust and reproducible*. London Academic Press.

Appendix:

Appendix A:

Appendix A.1:

Dependent Variable			
Name	**Type**	**Description**	**Reason**
Sales_volume	Int	Number of sales made by BMW	sales volume illustrates the number of customers buying that specific BMW car
Independent Variable			
BMW_Model	Categorical	Grouped category (Series, I, M, X)	Simplified models' classification for clearer analysis and visualisation
Model	Categorical	Specific type of BMW model (X5, M3,i8, etc.)	Identifies detailed product-level performance
Engine_Type	Categorical	Type of engine used in the vehical	Customer aesthetic preference
Fuel_Type	Categorical	The fuel type that the vehicle used	Customer aesthetic preference
Color	Categorical	Vehicle exterior decoration	Customer aesthetic preference
Transmission	Categorical	Is the car run automatically or manual	Reflect driving preference and convenience
Price_USD	Float	Vehicle selling price	Capturing the affordability and value perception
Region	Categorical	Geographic market	Use to identifies different market demand

In this analysis, “Sales_Volume” is selected as the dependent variable because it depicts the level of market demand for each BMW vehicle. Other variables include: “Fuel_Type”, “Color”, “Model”, “Price”, and “Region” which represent key attributes that may influence consumer purchasing decisions in the automotive market. Therefore, these variables are considered

independent variables and will be used to examine how different factors may affect vehicle sales performance.

Appendix A.2: Data Cleaning Process

The dataset itself is relatively clean and does not contain missing values or significant data inconsistencies. Initial checks were conducted to confirm data completeness and found no presence of duplicate records. However, all variables are set to general as default types, which are required to be converted into categorical variables. While variables such as Price_USD need to be converted to numerical variables as currency.

Appendix B:

Appendix B.1:

Sales Volume by Region				
Region	**Min of Sales_Volume**	**Max of Sales_Volume**	**Average of Sales_Volume**	**Sum of Sales_Volume**
Africa	101	9999	5036	41565252
Asia	105	9998	5083	42974277
Europe	100	9999	5106	42555138
Middle East	102	9998	5055	42326620
North America	100	9999	5087	42402629
South America	100	9999	5036	41551818
Grand Total	**100**	**9999**	**5068**	**253375734**

Table 2 summarise BMW sales performance across different regions. The results illustrate that the BMW average sales volume is relatively similar across regions. Although Europe has the highest average sales volume, Africa and South America present the lowest average. However, the total sales volume has reached its peak in Asia. Which suggests that BMW maintains a sustainable demand among different regional markets.

Appendix B.2

Sales Volume by Fuel Type				
Fuel Type	**Min of Sales_Volume**	**Max of Sales_Volume**	**Average of Sales_Volume**	**Sum of Sales_Volume**
Diesel	100	9999	5085	62361818
Electric	100	9999	5064	63157665
Hybrid	100	9998	5075	64532097
Petrol	102	9999	5046	63324154
Grand Total	**100**	**9999**	**5068**	**253375734**

Table 3 examines the BMW sales volume based on different fuel types. A consistent average sales volume across categories is displayed, ranging from approximately 5046 to 5085 units. Vehicles using diesel appeared to have the most average sales volume, while petrol vehicles demonstrated slightly lower averages, indicating an equal demand among different fuel types.

Appendix C:

Appendix C.1: Data Wrangling

During the data wrangling stage, one-hot encoding was applied to categorical variables to prepare for analysis. Likewise, a new variable BMW_Model was created from grouping a detailed variable into a broader variable for aiding visualisation in the later stage (Andrienko & Andrienko, 2010). On the other hand, unnecessary variables such as Sales_classification, Engine_size_L were dropped.

Appendix C.2:

```
bmw_df.describe()
```

	Year	Price_USD	Sales_Volume
count	50000.000000	50000.000000	50000.000000
mean	2017.015700	75034.600900	5067.514680
std	4.324459	25998.248882	2856.767125
min	2010.000000	30000.000000	100.000000
25%	2013.000000	52434.750000	2588.000000
50%	2017.000000	75011.500000	5087.000000
75%	2021.000000	97628.250000	7537.250000
max	2024.000000	119998.000000	9999.000000

Appendix C.3:

```
import pandas as pd
import numpy as np

bmw_df = pd.read_csv('BMW sales data.csv')
for col in ['Model', 'Region', 'Color', 'Fuel_Type', 'Transmission', 'Sales_Classification']:
    bmw_df[col] = bmw_df[col].astype('category')

bmw_df['Price_USD'] = bmw_df['Price_USD'].astype(float)
bmw_df['Price_USD'] = bmw_df['Price_USD'].round(2)

most_popular_car = bmw_df.groupby(['Model', 'Transmission', 'Color', 'Fuel_Type'])['Sales_Volume'].sum().idxmax()
max_sales_volume = bmw_df.groupby(['Model', 'Transmission', 'Color', 'Fuel_Type'])['Sales_Volume'].sum().max()
least_popular_car = bmw_df.groupby(['Model', 'Transmission', 'Color', 'Fuel_Type'])['Sales_Volume'].sum().idxmin()
min_sales_volume = bmw_df.groupby(['Model', 'Transmission', 'Color', 'Fuel_Type'])['Sales_Volume'].sum().min()

print(f"The least popular BMW car configuration is: {least_popular_car}")
print(f"With a total sales volume of: {min_sales_volume}")

print(f"The most popular BMW car configuration is: {most_popular_car}")
print(f"With a total sales volume of: {max_sales_volume}")
```

```
The least popular BMW car configuration is: ('X1', 'Manual', 'Grey', 'Electric')
With a total sales volume of: 313501
The most popular BMW car configuration is: ('i3', 'Manual', 'Red', 'Diesel')
With a total sales volume of: 713720
```

The prompts above are for extracting information on the highest and lowest sales of a specific BMW car to explore the popularity of the car. The results suggest that when all of the attributes are combined together, they suggest most people will buy a red i3 diesel BMW with manual transmission. While a grey X1 electric BMW with manual transmission will be less likely to be purchased by a consumer.

Appendix C.4:

```
bmw_df['Region'].value_counts()
```

Region	count
Asia	8454
Middle East	8373
North America	8335
Europe	8334
Africa	8253
South America	8251

Appendix C.5:

```
bmw_df['Fuel_Type'].value_counts()
```

Fuel_Type	count
Hybrid	12716
Petrol	12550
Electric	12471
Diesel	12263

Appendix C.6:

```
bmw_df['Transmission'].value_counts()
```

Transmission	count
Manual	25154
Automatic	24846

Appendix C.7:

```
bmw_df.groupby('Fuel_Type')['Price_USD'].mean()
```

```
/tmp/ipykernel_235/742208430.py:1: FutureWarning
  bmw_df.groupby('Fuel_Type')['Price_USD'].mean(
```

Fuel_Type	Price_USD
Diesel	75079.809671
Electric	75276.313207
Hybrid	74797.551746
Petrol	74990.419841

Appendix D:

Appendix D.1:

```
                            OLS Regression Results
...  ==============================================================================
     Dep. Variable:           Sales_Volume   R-squared:                       0.000
     Model:                          OLS   Adj. R-squared:                 -0.000
     Method:               Least Squares   F-statistic:                    0.5622
     Date:              Sun, 22 Mar 2026   Prob (F-statistic):              0.945
     Time:                      09:14:53   Log-Likelihood:            -4.6881e+05
     No. Observations:             50000   AIC:                         9.377e+05
     Df Residuals:                 49978   BIC:                         9.379e+05
     Df Model:                        21
     Covariance Type:            nonrobust
     ===============================================================================================
                                  coef    std err          t      P>|t|      [0.025      0.975]
     -----------------------------------------------------------------------------------------------
     const                   2833.5666   5961.083      0.475      0.635   -8850.225    1.45e+04
     Year                       1.1048      2.955      0.374      0.709      -4.687       6.897
     Price_USD              -3.65e-06      0.000     -0.007      0.994      -0.001       0.001
     Model_5 Series           -35.1512     59.624     -0.590      0.555    -152.015      81.712
     Model_7 Series            32.6844     59.388      0.550      0.582     -83.716     149.085
     Model_M3                  -0.6719     60.223     -0.011      0.991    -118.710     117.367
     Model_M5                  21.6263     60.005      0.360      0.719     -95.985     139.238
     Model_X1                  56.2753     59.694      0.943      0.346     -60.725     173.275
     Model_X3                  -7.4667     59.933     -0.125      0.901    -124.937     110.003
     Model_X5                  -3.2740     59.977     -0.055      0.956    -120.830     114.282
     Model_X6                  -3.8145     60.011     -0.064      0.949    -121.437     113.808
     Model_i3                 -55.3247     59.541     -0.929      0.353    -172.027      61.377
     Model_i8                  20.0297     59.575      0.336      0.737     -96.737     136.797
     Color_Blue                39.1822     44.444      0.882      0.378     -47.928     126.293
     Color_Grey               -10.5352     44.331     -0.238      0.812     -97.424      76.354
     Color_Red                  9.8446     44.177      0.223      0.824     -76.742      96.431
     Color_Silver              68.6329     44.328      1.548      0.122     -18.251     155.516
     Color_White               48.9261     44.391      1.102      0.270     -38.081     135.933
     Fuel_Type_Electric       -21.6623     36.339     -0.596      0.551     -92.887      49.563
     Fuel_Type_Hybrid         -11.3806     36.171     -0.315      0.753     -82.277      59.516
     Fuel_Type_Petrol         -39.8707     36.282     -1.099      0.272    -110.984      31.242
     Transmission_Manual       -8.2174     25.559     -0.322      0.748     -58.313      41.878
     ==============================================================================
     Omnibus:                    42496.722   Durbin-Watson:                   1.997
     Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2979.445
     Skew:                          -0.012   Prob(JB):                         0.00
     Kurtosis:                       1.804   Cond. No.                     3.71e+07
     ==============================================================================
```

Appendix D.2:

```
import matplotlib.pyplot as plt
import seaborn as sns

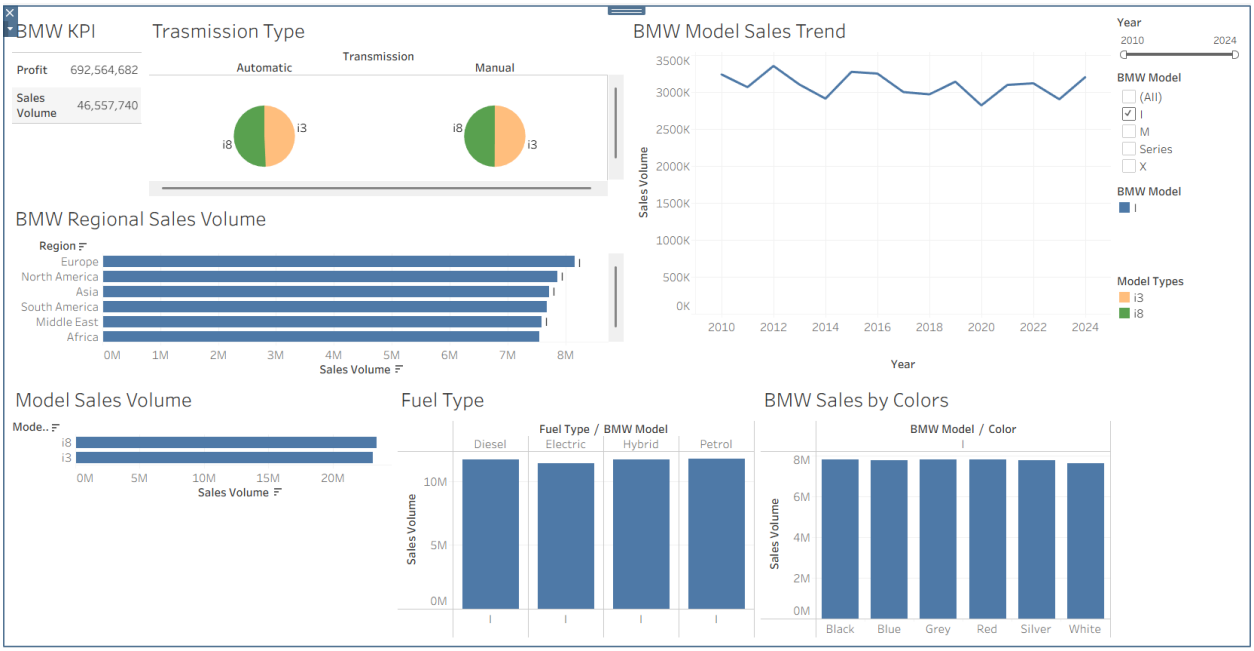
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Price_USD', y='Sales_Volume', data=bmw_df, alpha=0.6)
plt.title('Scatter Plot of Price_USD vs. Sales_Volume')
plt.xlabel('Price (USD)')
plt.ylabel('Sales Volume')
plt.grid(True)
plt.show()
```



The generated scatter plot displays the relationship between Price_USD and Sales_Volume. The data are fully spread over the entire graph, indicating a non-existing correlation between prices and sales. Frankly, the pricing does not significantly influence BMW sales performance from this dataset.

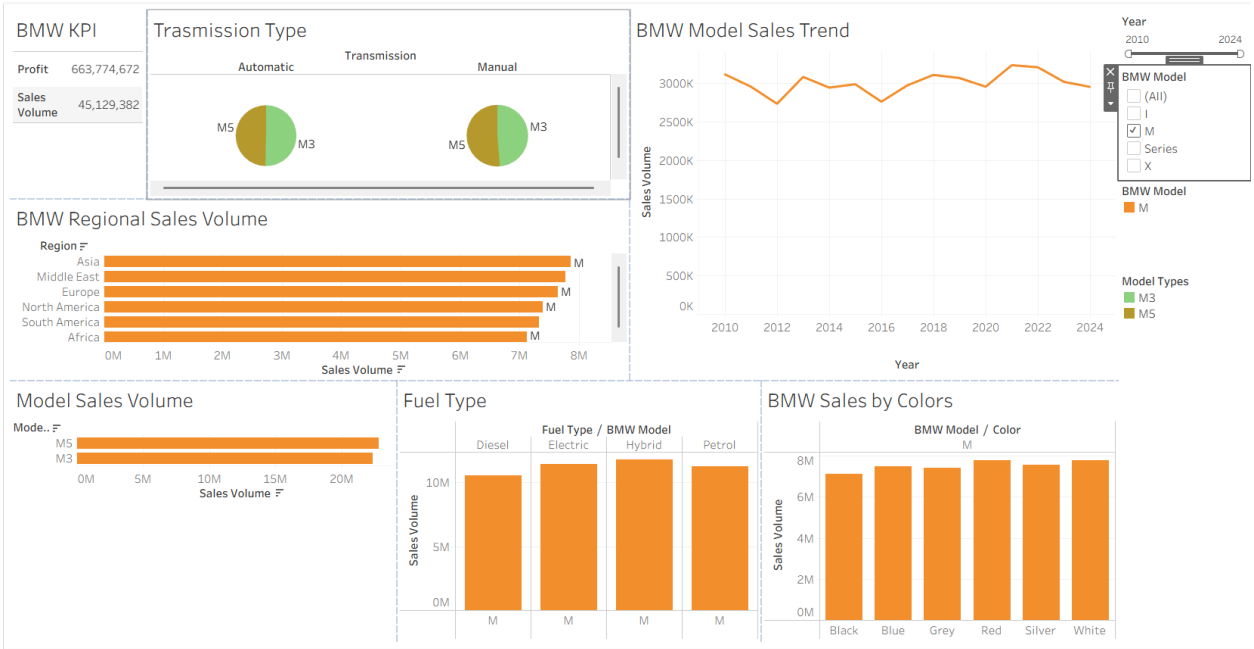
Appendix E:

Appendix E.1:



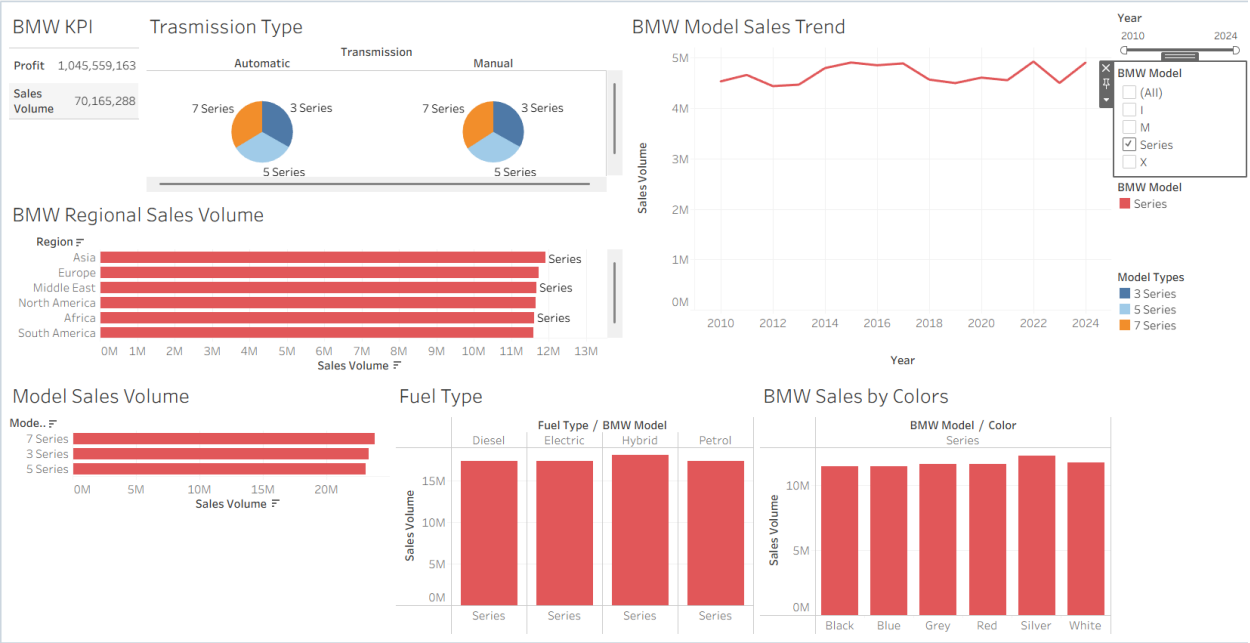
The I model experiences a volatile sales performance over time. Sales across fuel type, color, and transmission are equivalent, which indicates limited impact. Meanwhile, this specific model is the most popular in the European market.

Appendix E.2:



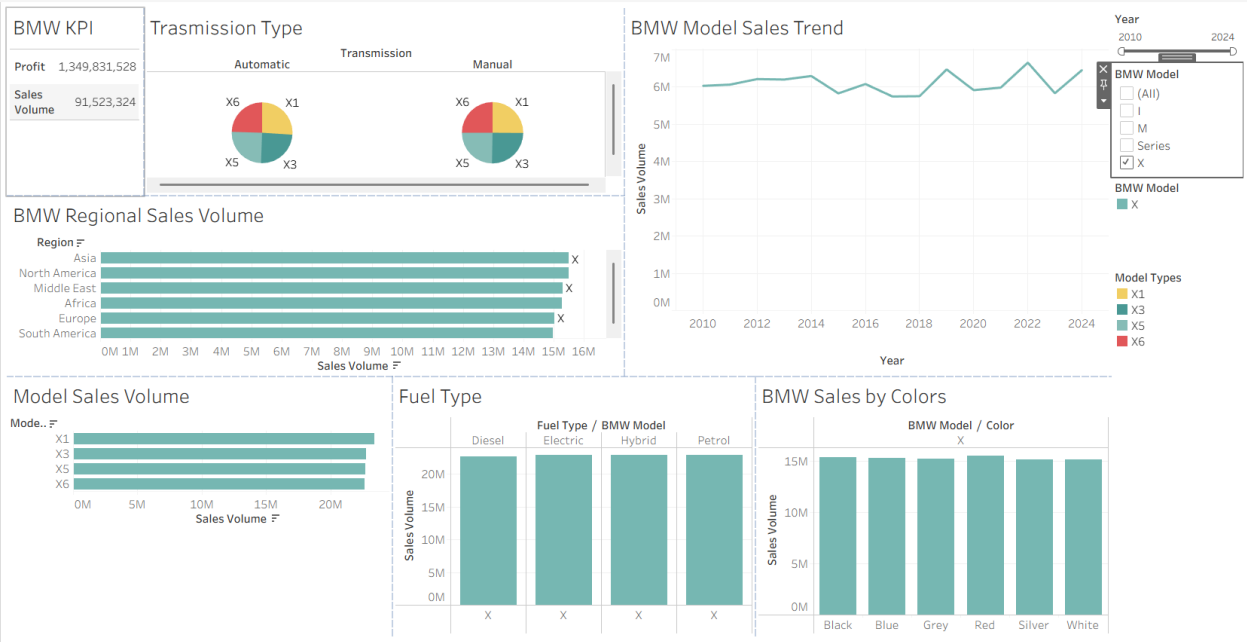
The M model generally displays lower sales volume compared to other models. Its sales trend presents a moderate fluctuation in sales with a slight decline toward the end. Asia remains the dominant market for this model. Furthermore, diesel variants and black color options appear to have slightly lower sales compared to other attributes.

Appendix E.3:



The series models demonstrate strong and consistent sales performance, with the 7 Series leading among individual models. Sales trends are relatively stable over time, with an increasing pattern. The sales volume is equally distributed across regions, indicating strong adoption in different markets.

Appendix E.4:



Model X has the highest overall sales volume, with a relatively stable trend in the first half of the period and becoming more fluctuating toward the end, alongside slight growth. Sales are consistent across fuel types, colors, and transmissions, suggesting no strong preference for specific attributes of this model. While Asia remains the dominant market region.